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Secure Peer-to-Peer Payment System on Android-Based Mobile Devices via Vibration and Acceleration

(Volume ___ of ___)

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ABSTRACT

The rapid growth of smartphone penetration in the modern era has enabled users to use mobile payment services for everyday transactions. The recent COVID-19 pandemic has significantly increased mobile payment usage and encouraged new account creation and mobile transactions in various peer-to-peer (P2P) mobile payment systems, such as Octopus and mobile wallets. The P2P mobile payment system enables two parties to securely transact by interacting in close physical proximity with mobile devices. The close physical proximity interaction is enabled by the Radio-frequency (RF) technology through Near Field Communication (NFC) and Bluetooth modules embedded in mobile phones. However, RF technology is vulnerable to several cyberattacks, such as eavesdropping and Man-in-the-Middle (MITM) attacks. It highlights the security problems of RF technology and encourages research on alternative solutions for P2P communication. The project proposes a new radio-free P2P communication channel based on vibration for mobile payment systems. The built-in vibrator and accelerometer on the mobile phone are used to send and receive vibration data, ensuring payments are conducted within a physical proximity. Moreover, to fulfill the security goals of the mobile payment system, such as confidentiality, integrity, and availability (CIA) measures, it is essential to integrate cryptographic frameworks for protecting the communication channel during transactions. However, vibration communication channels are low-bandwidth, making them unsuitable for traditional cryptographic protocols. In this study, we propose a lightweight cryptography (LWC) framework that incorporates out-of-band key establishment via a passphrase and PBKDF2, and integrates AES-CTR encryption with HMAC-SHA256 in the communication channel. These LWC protocols aim to ensure the transaction is safe. Finally, functional, Performance, and Penetration testing of the proposed secure vibration communication channel were conducted, which shows that it is feasible in a P2P mobile payment system under certain circumstances. However, the system has demonstrated an alternative, secure, radio-free communication methodology that can potentially be used in further research on P2P communication and lightweight cryptography under a payment system.

Keywords—*P2P Payment System, Vibration-based Communication, Mobile System Security, Lightweight Cryptography*

1 Introduction

This section will begin with a summary of revisions and additions made after the submission of Interim Report I. Next, it will introduce the background and research gap. Moreover, it will also define the project objectives, justify the scope, and outline the deliverables for developing a secure Peer-to-Peer (P2P) payment system using a physical vibration channel. Lastly, it will provide an overview of the paper's organization.

1.1 Summary of revisions and additions since Interim Report I

In Interim Report I, my supervisor commented that the background discussion could be more extensive and detailed, including additional examples of related methods. He also expected that Interim Report II would include the full design and implementation of the proposed system, along with initial results.

Therefore, in this report, I have added more detailed examples, such as discussing the implementation of Visual Light Communication and a high-frequency acoustic wave channel in Section 2.1.3, and comparing it with the proposed vibration channel. Moreover, I have included the detailed overall design and implementation of the proposed system in sections 3 and 4. Finally, the preliminary results and future improvement are also incorporated in Section 5.

1.2 Background of the Study

The increasing use of smartphones has fundamentally transformed the financial structure of modern society. Peer-to-Peer (P2P) payment systems have appeared as a keystone of modern digital payment transactions, enabling money transfer between users via mobile devices directly without using cash [1]. The recent COVID-19 pandemic has increased the acceptance of mobile payments, and the P2P payment market proportions reached $1,889.16 billion in 2020 and are valued at approximately $9,097.06 billion by 2030 globally [2]. Currently, Hong Kong has the highest percentage (46.7%) of interested users who use mobile payment system services monthly in the Asia-Pacific [3]. The P2P payment systems are believed to skyrocket and play an essential role in digital financial services worldwide.

Radio-frequency (RF) technology has seen rapid advancements in wireless network communication and mobile technology. Wireless payments using technologies such as Near Field Communication (NFC) and Bluetooth have become popular and are a standard protocol for short-range wireless communication channels. NFC has been adopted in P2P payment systems like

Apple Pay, PayPal and Octopus [4, 5]. However, the RF-based approach requires particular hardware and can be vulnerable to compromise, such as eavesdropping and relay attacks [6]. Therefore, RF's drawbacks have encouraged the use of different channels for P2P communication, such as the High-Frequency Acoustic Wave channel [7] and the Visible Light channel [8]. Mobile phone device vibration and accelerometer sensing have also been proposed as a communication channel [9, 10]. These approaches may enable an offline physical P2P payment system to function without radio frequency.

1.3 Problem Statement

Previous research has successfully demonstrated the feasibility of using a smartphone's vibrator and accelerometer to establish data transfer through a physical channel [9, 10]. However, these studies have primarily focused on basic P2P communication for experimental purposes. Implementing the physical vibration channel, particularly in financial applications such as a P2P payment system, remains largely unexplored. Few studies have examined the feasibility of the offline physical vibration-based P2P payment system. Proposing an innovative method for future research is necessary.

Moreover, deploying a security solution like an encryption scheme is crucial for protecting the P2P payment system against threats. However, the use of high-overhead traditional cryptographic protocols is not suitable in the case of vibration-based communication with severe constraints. Scholars have found that asymmetric encryption and public-key infrastructure (PKI) are particularly unsuitable. Their computational complexity drains resources, and a significant communication overhead is difficult for a low-bit-rate channel [11]. This creates a critical research gap in designing a lightweight cryptographic mechanism for high-latency, low-throughput vibration-based communication inside the P2P payment system.

1.4 Aim and Objectives

This project aims to tackle the insufficiency of using the vibration channel in the offline P2P payment system and design a lightweight encryption and key exchange mechanism for this communication channel. The objectives are as follows:

- To design a Vibration-Based Communication Protocol for the system: Develop an application-based algorithm for reliably encoding, transmitting, and decoding the data between two Android smartphones using the devices' vibrator and accelerometer.

- To develop a Lightweight Cryptographic Framework: Implement a simple security mechanism for the vibration channel. These include a secure custom method for secret key exchange and a simple encryption scheme for the transaction data.

- To evaluate the system performance and security: Test the final prototype of the application, and assess its performance, such as accuracy; Analyze its security against a described threat model and identify possible vulnerabilities and limitations.

1.5 Project Scope and Deliverables

This project will develop an application on Android devices that will only use the built-in vibrator and accelerometer for communication, excluding other technologies like Wi-Fi or Bluetooth. The project outcome will be a functional P2P payment System with the proposed communication protocol and cryptographic framework. The scope does not include development for other operating systems (OS) and defense against advanced physical attacks.

1.6 Overview of the Interim Report II

The rest of the paper is organized as follows. Section 2 will conduct a literature review to indicate the necessity of the work. Section 3 presents the overall design of the proposed vibration-based P2P payment system. Section 4 illustrates the detailed implementation within the system. The final section presents the preliminary results and outlines further improvements to the system.

2 Literature Review

This section begins with an overview of existing P2P mobile payment system solutions, including NFC and Bluetooth, and a review and comparison of alternative solutions to overcome the weaknesses of the existing P2P solutions. Next, a review of current vibration-based communication channels will be conducted, assessing their application and inadequacies for this project. Additionally, it will provide an evaluation of a cryptographic framework for vibration channels, followed by a critical examination of its limitations in the system.

2.1 Existing P2P Payment Technologies and Security Challenges

2.1.1 Near Field Communication (NFC) & Security Vulnerabilities

Nowadays, most P2P payment systems adopt the use of Radio-Frequency (RF)-based technology like NFC in personal electronic wallets (e-wallets) for contactless payment [13, 14]. NFC is a short-range P2P communication tool with radio frequency that combines Radio Frequency Identification (RFID) benefits with wireless communication technology like Wi-Fi. NFC is being introduced as a simple operation technology, as users only need to wave the mobile device close to another device to receive NFC signals, and the transaction can be completed immediately without other actions required [15]. This has fostered convenience and flexibility by tapping mobile devices for P2P payment without typing transaction details [16]. However, the protocol introduced a susceptibility to specific vulnerabilities, such as eavesdropping and Man-In-The-Middle Attack (MITM Attack), due to dependency on continuous internet connectivity [6, 17].

Regarding eavesdropping, NFC faces significant security challenges that create opportunities for alternative approaches. Previous research has shown that data on the NFC-enabled device can be intercepted at a distance by an attacker who listens to the communication via RF waves that transmit personal information or payment details and extracts the data from received RF signals during a P2P payment transaction. [18, 19]. This issue highlights security and privacy risks. Data branching can lead to unauthorized access, allowing attackers to gain unpermitted access to users' financial data throughout the NFC process. This violates the security principle of availability of "unauthorized denial of use" as the attackers can use and access the information without permission [20, 21]. Moreover, the data exposure risk has caused the attackers to collect and track users' information, and potentially utilize it wrongly to commit a crime or create a duplicate card that contains credit card data for card skimming [22].

Concerning the MITM attack, it could occur during NFC payments. Some researchers found that some NFC devices are connected to the cloud (Internet) channel instead of the traditional NFC, where cryptographic operation happens inside the tamper-resistant hardware (Controller) to prevent cyber attacks [23]. They discover that if the communication channel between the mobile device and the cloud is not adequately secured, such as weak HTTPS protocols or unverified certificates, the attackers may intercept the traffic by creating untrusted CA certificates that fool users into signing them. and keep the MITM proxy for accessing the information during the transaction [23, 24].

The security loopholes of NFC highlight the need for an offline-capable payment solution that can operate without continuous internet dependency and radio-free communication protocols to avoid possible attacks.

2.1.2 Bluetooth & Security Vulnerabilities

Apart from NFC, Bluetooth is also a radio frequency (RF)-based technology that interconnects mobile devices with each other via P2P communication. Currently, the latest standard available is versions 5 and 6 with Bluetooth Low Energy (BLE), which helps to reduce the power consumption cost while maintaining short-distance communication [25]. Bluetooth operates on radio frequencies in the 2.45 GHz band, with a transfer rate from 1Mbps to 2 Mbps, and a transmission range from 10m to around 240m in omnidirectional format [25, 26]. Some researchers have implemented a “P2P-Paid system” that successfully conducts P2P mobile payment transactions on mobile phones via a nearby Bluetooth wireless network, showing the possibility of providing a functional P2P payment system using Bluetooth [4]. However, similar to NFC, Bluetooth relies on an RF signal that causes Confidentiality, Integrity, and Availability (CIA) security problems [27].

Regarding the Bluetooth security issues, previous research found that Bluetooth-enabled attacks are prone to security problems. For example, a Bluetooth sniffer can intercept the raw data being exchanged between two mobile devices [28], which could cause a serious problem with the confidentiality of transaction information since third parties may collect the sensitive data without notice. Furthermore, *Figure 2.1* shows that a MAC Spoofing attack can be made before the Bluetooth pairing between mobile devices. The attackers can clone a MAC address of the original device and impersonate it to manipulate data while in transition, showing that the information is being altered to confuse the receiver, violating the integrity [27, 29].

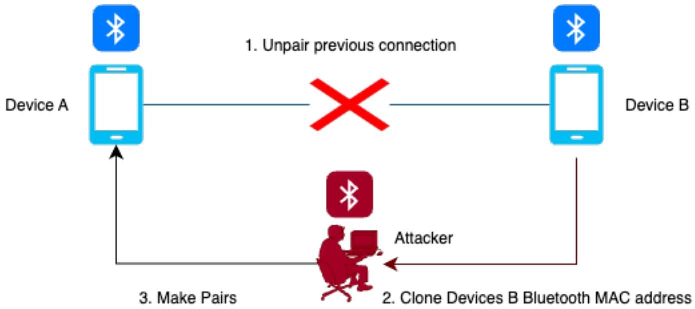

Figure 2.1: A MAC spoofing attack scenario using Bluetooth

Additionally, Bluetooth transmission must transfer in the form of RF waves. An attacker introduces sufficient RF noise into the Bluetooth network. Unwilling echoes can overwhelm the connection and prevent the establishment of communication [28]. This represents a Denial of Service (DoS) attack that blocks users from accessing a service by making it unavailable to users, which violates the availability principle in security [29].

The security inadequacies of Bluetooth emphasize the ongoing challenges in wireless P2P payment technology. These limitations create opportunities for a more secure and offline option, a P2P communication protocol in a payment system, like vibration-based solutions.

2.1.3 Secure Offline Covert P2P Communication technologies

To overcome the limitations of NFC and Bluetooth, research on alternative solutions such as physical covert P2P communication channels has demonstrated that they offer better physical security. The radio-free method needs close physical contact between mobile devices making it resistant to wireless attacks, such as eavesdropping [10]. Additionally, the lack of address discovery via the network or devices, as well as the short communication range, further safeguards data transfer [30]. Currently, there are some covert channels, for example Visual Light Communication (VLC) with Light Emitting Diodes (LED) approach proposed by Schmid et al. [8], where the approach sends information by a standard LED driven by a designed modulator that changes light intensity at a data rate to send out messages, and the receiver operates the LED in a photodetection mode that converts detected light into a photocurrent and decode it to a message. Additionally, Li et al. have proposed a communication channel based on high-frequency acoustic waves [7]. In which the sender plays high-frequency sound wave data through the air medium, then the receiver captures the audio and decodes the data. Moreover, many scholars have proposed the vibration communication channels [9, 10, 31, 33, 34, 35]. The data transmission sends a vibration pattern that encodes data using the smartphone's built-in vibrator, which is then decoded by receiver using the smartphone's accelerometer. In the context of the P2P mobile payment system, vibration will be a more suitable methodology compared to other covert channels. Compared to the LED approaches, the LED requires additional hardware modules such as an Arduino board and LEDs embedded in the mobile device, which can be a burden for carrying and not user-friendly for mobile users, and may also highly interfere with environmental factors (Such as sunlight or indoor lighting) during transactions [8]. Compared with the acoustic approach, although acoustic signals can be implemented through built-in smartphones' microphones and speakers, from an interference perspective, vibrations are less vulnerable to ambient noise, as busy or even quiet acoustic environments may degrade the sound channel [7]. The vibration on a table is relatively isolated from noise and light, showing the channel is separated from the background environment, and it is a user-observable channel with strong physical proximity binding that helps to reduce the risk of possible conversion attacks[9, 10].

Therefore, to develop reliability and secure payment without connectivity requirements, this project proposed a sustainable vibration-based P2P transaction solution that ensures financial inclusion for those concerned about payment privacy and security.

2.2 Vibration-Based P2P Communication Technology

2.2.1 Vibration and Acceleration Data Transfer in Mobile Devices

Vibration-based P2P Communication is a non-radio-based, low-rate, and short-range communication that enables users to transfer data among smartphones by just putting one smartphone on another, similar to NFC communication [10, 31]. Communication requires the raw sensor data of the accelerometers and vibrators for the information sender and receiver, and this is feasible since this type of motion sensor is provided by the Android Hardware [32]. The data transmission uses a built-in vibrator inside the smartphone to send a vibration pattern that encodes data decoded by the smartphone's accelerometer as a receiver. Current Vibration Communication methodologies adopt a similar decode mechanism using simple binary and different adaptive thresholding; however, they differ in the encoding phase [33]. There are three primary encoding methodologies: On-Off Keying (OOK)-based protocol [10], Amplitude modulation (AM)-based protocol [34], and Frequency Modulation (FM)-based protocol [35]. Previous research shows that OOK is considered more practical when applied to mobile devices, as it is simple to implement in all mobile phones by just switching the vibrator on/off to represent 1s and 0s using basic motors and API without considering the vibrator strength or frequency that may require additional hardware or OS support [33]. Therefore, OOK is the most suitable method for mobile devices, considering the security and convenience of development.

2.2.2 Applications and Limitations in Previous Approaches

In the context of P2P communication, several research projects use the mobile phone vibrator and accelerometer for data transmission. Hwang et al. proposed VibComm, an offline Android application that uses vibration-based communication to exchange data between multiple users [9]. Yonezawa et al. have also developed Vinteraction, an information-transmitting application that leverages a combination of an accelerometer and vibrator using smartphones [31]. Both previous studies used vibration for P2P communication with OOK encoding and a similar adaptive thresholding decode mechanism, and it is offline and radio-free. Results found that 10cm is the minimum distance between two mobile phones, showing that data was successfully sent to other devices. However, they found that the transmission rate is low, limited to 0.09 packets or 10 bits per second, since a higher transmission rate will affect decoding accuracy [9, 31].

Roy and Choudhury have further proposed Ripple I and II [34, 35]. They have tried implementing AM and FM transmission protocols using a customized vibrator and accelerometer. Results show that they have pushed that transmission rate to 30 - 200 bits/s, and it is protected against the eavesdropper [34, 35]. However, it requires higher-level and self-developed mobile phone modules, which are impossible for ordinary users to develop. Finally, Zhao et al. have proposed LeaD [33]. They have modified the OOK encoding scheme to a bit group and decoded the bits using machine learning (ML) models for classification. The best ML model can boost the transmission rate to 40 bits/s with high accuracy [33].

The above scholars have inspired this project to adopt a Vibration-based scheme for P2P communication. However, previous studies have primarily focused on establishing a basic communication connection and transmitting simple strings for experimental purposes only. They often assume that the channel itself is a trusted and safe one. Implementing a secure protocol for the physical vibration channel remains largely unexplored, particularly in financial applications such as a P2P payment system. The project also hopes to bridge this gap and show the possibility and feasibility of the security vibration-based P2P payment system.

2.3 Cryptography Framework for Vibration Channels

2.3.1 Security for P2P Mobile Payment Systems

P2P Mobile payment system security is essential for the users and service providers. The transaction data must be protected when it is used. The desired security goals in a mobile payment system include integrity, confidentiality, availability, authentication, access control, and non-repudiation [36].

To illustrate more, confidentiality requires that the data be restricted only to the parties involved in the transaction process and protection for passive attacks [37, 38]. Integrity ensures the system's accuracy during data processing, which is not being modified in a complete and timely manner [37, 38]. Moreover, non-repudiation ensures that it is a specific user’s transfer transaction messages and prevents them from denying the services [38, 39]. Availability ensures the mobile payment system is accessible upon the user's request [38, 39]. Next, authentication requires verifying the users and the data origins [38, 39]. Finally, access control ensures only authorized persons can access the mobile payment system and blocks unauthorized personnel from accessing the payment system [38, 39]. This highlights the significance of a secure P2P mobile payment and the cryptographic framework, such as Symmetric and Asymmetric encryption, PKI, and digital signatures. These provide foundational tools to safeguard the mobile payment system and its security goals.

However, cryptographic challenges will be found in the low-bandwidth environment due to different constraints, such as a vibration channel. The sub-section below will discuss the relevant issues.

2.3.2 Traditional Cryptographic Challenges in Low-Bandwidth Environments

As mentioned in section 2.2.2, the vibration-based P2P communication only has a 10-200 bits/s transmission rate, which is significantly slower compared to other P2P communications used in mobile payment systems like Bluetooth with 1-2Mbps (Mentioned in 2.21) and NFC with up to 424Kbps [40]. Moreover, Mobile devices often work under constraints like limited processing power, memory, and battery capacity [41]. The twofold challenge has created a Low-Bandwidth and resource environment for the P2P payment system using vibration.

However, using high-overhead traditional cryptographic protocols is unsuitable in the case of a vibration P2P payment system, as it is difficult to process both vibration communication and cryptography in parallel. For example, previous research indicated that using asymmetric encryption and PKI in vibration-based channel devices leads to resource drainage due to computational complexity and significant communication overhead challenges for low-bit-rate channels [11]. Moreover, running a conventional cryptographic algorithm on a mobile platform consumes many resources, like CPU time and memory, making battery power drain fast and transactions slower [42, 43].

Yet, cryptography is necessary for establishing a secure mobile payment system. It must not be omitted to achieve the security objectives. Therefore, Lightweight Cryptographic (LWC) algorithms are essential in mobile payment systems to balance security and performance. The LWC framework provides strong security properties while minimizing the computational power to ensure a user experience and transaction time.

2.3.3 Effective Lightweight Cryptographic (LWC) Framework

The LWC Framework is not a new concept in cryptography. DES Lightweight extension (DESL) was first introduced in 2007 as a lightweight version of DES encryption, and the idea is simple: take DES and lighten it so that mobile devices can perform encryption at a low cost [44]. All the LWC shares common goals to run light and perform security goals, no matter how the algorithm is made. As Johnny pointed out, the LWC is feasible on mobile devices, and many lightweight API cryptography packages are supported for the mobile application environment [45]. At the same time, many LWC frameworks are supported. Our studies will focus on reviewing the algorithm used in our project.

Previous research has shown that symmetric cryptography is one of the lightweight solutions for the vibration communication channel [11]. Symmetric algorithms are generally faster and require less computational power than asymmetric ones, making them ideal for devices and communication channels with limited processing power, like smartphone motion sensors [46]. Advanced Encryption Standard (AES) is a standard symmetric LWC that provides security from a confidentiality and authentication perspective. AES is a block cipher that uses a fixed-length key and block sizes for encryption and decryption, and offers strong security guarantees with key sizes of 128 to 256 bits. Moreover, AES is part of many international and regional standards, which makes it a default option for secure communication in constrained environments [46]. From a lightweight perspective, many smartphones currently include an AES hardware module that allows encryption to be executed without burdening the CPU and memory. In symmetric cryptography, the increase in key sizes does not significantly increase energy consumption, as the key expansion step in AES is only a small part of the total computation [46, 47]. The results concluded that AES may be suitable for the vibration communication channel, such as the LWC.

However, we have to consider the constraints of the AES approach to encryption when implementing it in the studies. First, the block size of the data will be a challenge. As mentioned in the previous literature, the key size is fixed in relation to the block size; if we choose a larger key size for better security, the block size will also increase. However, 128-bit blocks can be excessive for devices that only transmit small messages [46]. We must consider the tradeoff between security, computation power, and time when designing an AES encryption scheme for the vibration channel. The possible solution lies not in changing AES, but in changing the operation mode, as the operation's use will affect smaller or larger messages in a single block. In traditional protocols like AES-Cipher Block Chaining (CBC), when the message size is smaller than the fixed block size, padding is triggered and extra data is added to fill the blocks [48]. The method is not ideal for a vibration communication channel, as the limited bandwidth is wasted to process the redundant data created by padding. One of the modes that is more suitable in a lightweight mindset is AES-CTR (AES-Counter). The approach converts the block cipher to a stream cipher, which utilizes a counter (or nonce) with a key to construct a key stream and XOR it with the plaintext to form the ciphertext for communication (*Figure 2.2* below shows a typical flow of CTR mode) [49, 50].

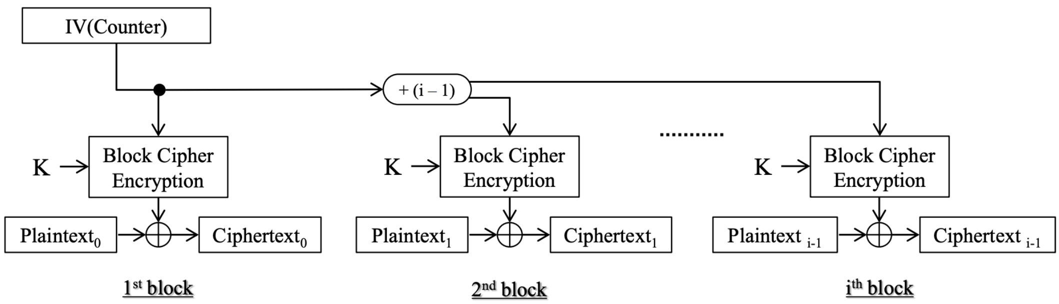

Figure 2.2: CTR Mode Encryption [50]

The CTR encryption shows that it is more parallelizable and doesn't require padding of the block while creating a ciphertext the same size as the small message being sent. It offers higher flexibility in determining the message size, which may suit a lightweight approach.

Moreover, Prior studies indicate that symmetric cryptography, which relies on exchanging a shared secret key between two devices, presents challenges in the vibration channel [11]. *Figure*

2.3 shows that symmetric key cryptography uses the same private key for encryption and decryption, and both users require a pre-shared secret key to be established between parties before they can communicate securely [51].

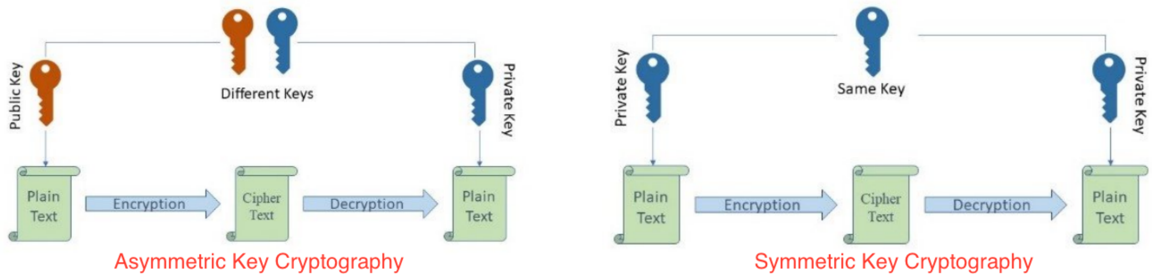

Figure 2.3: Asymmetric & Symmetric Key Cryptography processes comparison [51]

This presented the challenge of how to set up and exchange the key before communication in a lightweight method, as the energy consumption and resource demands of an AES depend not only on the process but also on the key setup cost [52].

A form of data transmission protection is password-based encryption, which involves using a password or passphrase input by the end users, and it serves as a cryptographic key for encryption [53]. It is an ideal lightweight methodology for key establishment since the protocol is designed for a few message exchanges and uses simple cryptography operations, which makes it suitable for a low-bandwidth channel [54]. Yet, it is very vulnerable from a security perspective since users will prefer weak passwords that are easy to remember; these passwords are generally short and easy to guess, like using their birthday as a password, which is highly susceptible to dictionary attacks that attackers try all possible passwords, until they find the correct one from a defined file [55, 56]. To reduce the weakness of the password, the "Password-Based Key Derivation Function version 2" (PBKDF2) has been introduced by the researcher to secure the password before use to establish a robust secret key. The PBKDF2 is denoted by: $DK = PBKDF2(p, s, c, dkLen)$.

The algorithm will finally output a derived key DK where p is the passphrase input by the user, a random salt s, an iterated loop counter c, and the secret key length $dkLen$ [57]. The PBKDF2 added salt to prevent rainbow table attacks and applied hashing like HMAC-SHA256 many times on a defined iteration loop that establishes the secret key for the AES-secured. It is also a suitable approach in a lightweight context since it is simple to implement. The key length and iteration count can be tuned based on mobile device capabilities, which provide considerable flexibility for customization [58, 59]. Password-based encryption with PBKDF2 is a suitable lightweight method for key setup and exchange in our project.

Lastly, we should consider a cryptographic protocol to protect authentication and integrity during message transfer. Message authentication codes (MAC) are a symmetric cryptographic mechanism that allows two users with the same shared key to authenticate the data transmitted. *Figure 2.4* illustrates the whole simple process of the MAC protocol. First, the sender computes a MAC tag using the message *M* and key *K*; then, it sends the message with the tag as a block to the destination receiver. Next, the receiver computes a new MAC tag with *M* and *K* and compares it with the MAC tag received. If it is equal, the receiver can conclude it is not altered during transmission [60].

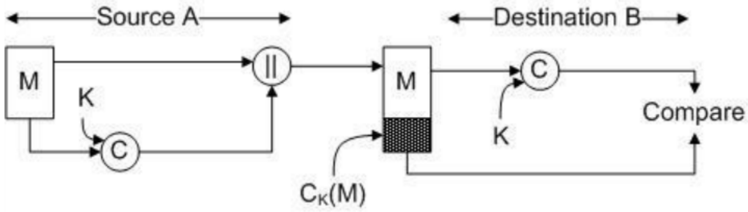

Figure 2.4: Overview of the Message Authentication Code [60]

From a security perspective, if we assume only the two parties know the secret key and the calculated MAC tag matches the received MAC tag, then:

- Integrity: The tag helps to verify that the message has not been modified since any alteration would cause verification to fail [60, 61].
- Authentication: The tag assures the message is from an authentic sender, proving that the data originated from a user who knows the secret key [60, 61].

The studies have proven that we should consider the MAC design when implementing a cryptographic framework. Nevertheless, given the limited processing power of the communication channel of vibration. We should also consider using a lightweight approach to the MAC. Previous studies have proposed a lightweight MAC using a “Hash-based MAC” (HMAC) approach to ensure security goals while keeping computation costs low in low-bandwidth channels, such as those found in IoT devices [62]. The simple structure of the HMAC is relatively clean, where the function is $t = H(K, M)$, where t is the MAC tag of the Message M and the key K, and a hash function H using a compression algorithm like SHA-256, and the complexity of the whole function depends on how the key and message are constructed [63]. Consider the design of lightweight HMACs by previous scholars, which only use hashing, the XOR operation, and concatenation of data. These are computationally inexpensive and reduce the communication overhead by minimizing the bits needed to transfer, but maintain the original security measures [62, 64].

To conclude, Section 2.3.3 provides a clear insight into LWC components to secure communication over constrained channels like vibration. Previous research supports the use of AES, out-of-band password-based encryption with PBKDF2, and the HMAC with a lightweight framework can help to fulfill the security goals of a P2P payment system. However, while these protocols are well-researched individually, there is a significant gap in integrating protocols within the vibration channel P2P payment mobile system. The project hopes to design and evaluate a lightweight cryptography framework for the secure P2P vibration payment system.

3 Proposed Design

This section provides an overview of the design for vibration-based communication in the P2P payment system, along with the software engineering (SE) approach considered. Next, it will also discuss how the system will be evaluated generally, including the testing procedures and setup. Lastly, the project's development cycle will be introduced.

3.1 Overall P2P Vibration Mobile Payment System

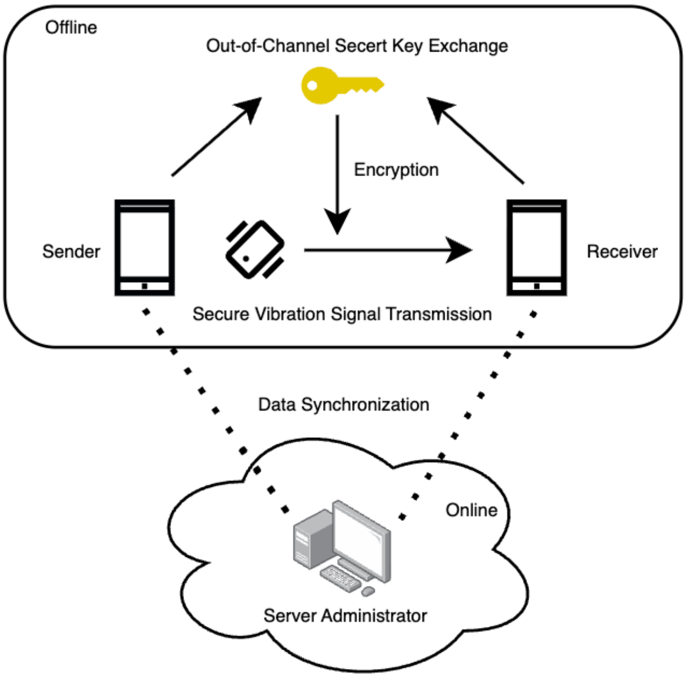

Figure 3.1: System diagram of the proposed solution

Figure 3.1 illustrates the overall system architecture of the vibration-based P2P payment system. The system will consist of three major components. The first components focus on the functionalities of the mobile payment system, including system features for users and the design of vibration communication. The second component concentrates on a payment management system for administrators, including transaction log recording and user data synchronization. The last component focuses on the lightweight cryptographic design, incorporating the secret key exchange and the encryption scheme for the vibration communication. All transactions can be completed in a physical and offline layer, except when data synchronization and transaction log saving are required from the server side.

Below are the overall technical framework and libraries used for this project:

For the coding environment, we will use Android Studio IDE and Kotlin to develop each component of the project.

For the UI and functionalities of the system, the project will utilize Jetpack Compose, an open-source toolkit for building native UI on Android.

For the vibration communication channel, the Android Haptic APIs will be used to generate and control the vibration channel, and the Android Sensor Framework (*TYPE_ACCELEROMETER & SensorEventListener*) will be utilized to receive and listen to the vibration data.

For the Lightweight Cryptographic Protocol, the project will apply javax.crypto API for cryptographic operations such as SecretKeyFactory and Mac.

For the database that stores user data and transaction records, a local database file will be used, as it serves as a test database and backend for various applications, including authentication, storage, and database management.

3.2 Functionalities of the Proposed System

Users can log in, log out, and create an account for the payment system. Moreover, they can also update their status, including their displayed username and passwords. This provides users with an authentication and customization function.

Moreover, users can check their payment history on the history screen, which will be categorized as “Pay” and “Receive payment” records. All payment logs will be sorted by date and time in descending order by default for improved display. Users can also search for specific transaction records by using keywords related to the transaction details.

Furthermore, users can send and receive payments from other users. If users choose “Send Payment”, the vibration encoder in the system will encode the vibration pattern of the payment based on the customized OOK algorithm and send it to the receiver for payment. If users choose “Receive Payment”, the vibration decoder in the system will listen to the vibration pattern from the transmitter and decode the payment vibration pattern using the adaptive threshold algorithm, and finally confirm the payment.

Prior to encoding or decoding the vibration payment pattern, the system requires both the transmitter and receiver to input a verbally agreed-upon secret passphrase. The system then generates a Pre-Shared Secret (PSS) key by processing this passphrase, which is used to encrypt and decrypt the vibration payment data.

In Section 4, this paper will introduce the detailed implementation of the vibration communication channel and the lightweight cryptographic protocol.

Lastly, if users are online (with an internet connection), they can update their account status and transaction log on the online server. From the server administrator's management side, they can check the transaction log and user status on the server.

Figure 3.2 below illustrates a use case diagram that describes the system's high-level functions and identifies the interactions between the system functions and its users.

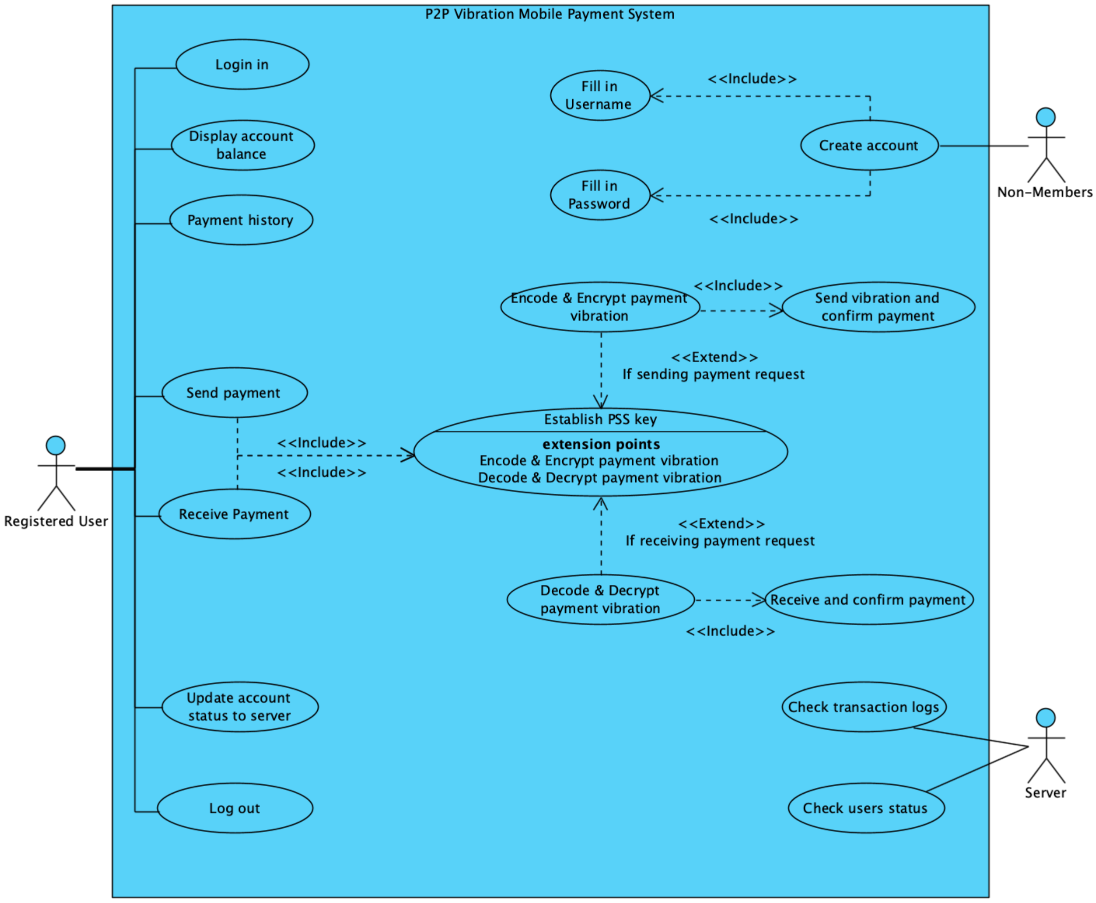

Figure 3.2: Use case diagram of the proposed solution

3.3 Software Engineering Approach of the Proposed System

We should also consider a software engineering (SE) approach, such as design principles and patterns, before implementing the whole system.

First, we should consider the design principles. The Single Responsibility Principle (SRP) ensures that every class or object should have only one job, which helps increase maintainability by making components highly reusable and easy to test. For example, vibration design should ensure that the vibration encoder is only responsible for encoding the payment request to a vibration pattern and the decoder should only perform the reconstruction of the original data of the vibration.

Furthermore, the Open-Closed Principle (OCP) should be considered: the system's functions should be open to extension and closed to modification. OCP helps minimize the impact on existing code when making changes and makes it easy to add new functions. Abstraction and subclassing techniques can help to achieve the principle. For example, the Lightweight Cryptographic Protocol always requires encryption and decryption of the message. Their used attribute and behavior will always remain the same, and the only difference is the cryptographic algorithm used in each process. Therefore, the OCP can ensure that the class is always open for extending the use of the protocol, while maintaining the behavior and attributes defined in the design.

Lastly, this project should also consider the KISS principle, which emphasizes keeping the project simple and reliable to solve complex problems. For example, in this project, we are evaluating the secure vibration channel but not developing a complex mobile payment system, so we should implement a simple function to demonstrate each use case using the proposed solution.

Next, we should consider the design pattern. The singleton pattern should be considered; this pattern ensures that only one instance of a class exists in the entire application. It provides better control as there is only one vibration controller for physical vibration, and prevents duplicated parts from performing the same tasks.

Additionally, the observer pattern is considered, where there is a listener is notified of changes from the publisher. In our project, the UI of each component should observe the update of the variable and update the display information automatically. For example, the home menu observes the changes in the user's balance and displays the actual balance of the user.

Lastly, to ensure a program performs tasks in the background without halting its main execution, it is necessary to prevent the vibration channel from requiring the user to idle the mobile phone for sending and receiving vibration patterns, as this would completely freeze the interaction between the application. Kotlin Coroutines are used for structured concurrency and thread confinement. For example, the send payment coroutine in HomeMenu runs in the background, but updates the transaction status concurrently to prevent memory crashes or loss.

Figure 3.3 illustrates a simple class diagram of the proposed design, which incorporates the considered design patterns and principles.

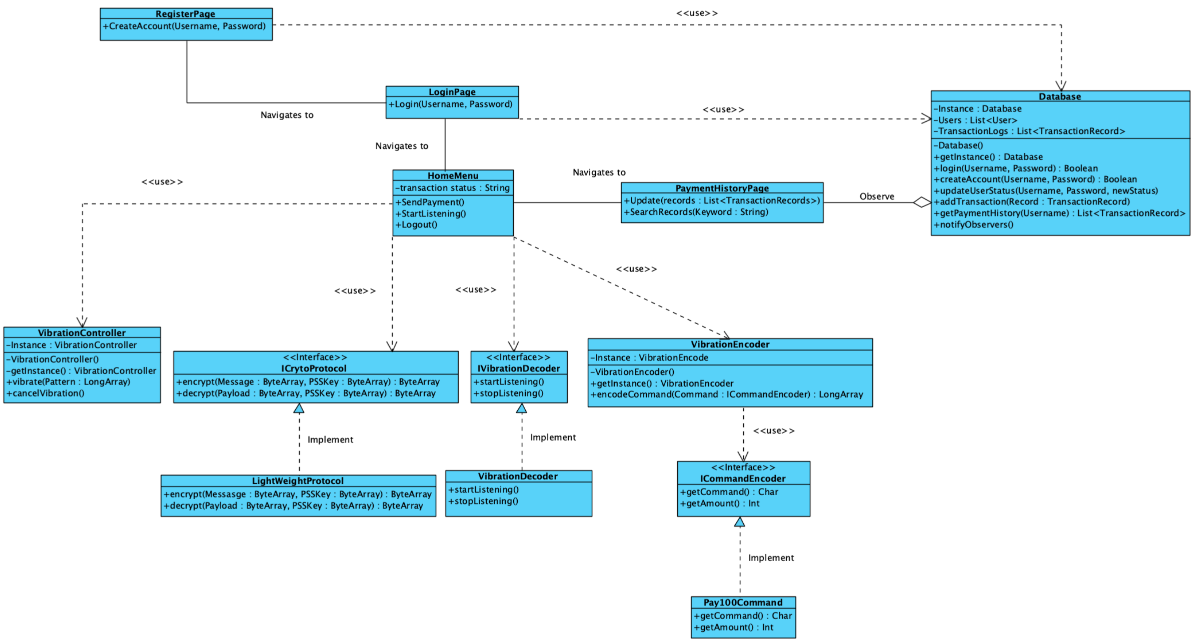

Figure 3.3: Class diagram of the proposed solution

3.4 Performance Evaluation Design

The final prototype will be tested on two Android phones. A Bottom-up approach will be used to test the system's functionality, followed by performance and penetration tests for testing the vibration algorithm and security protocols.

Unit Testing

During system development, we will perform unit tests for each developed class, following the functions specified in Section 3.2.

Integration Testing

After completing each phase, we will integrate the subset of each class into a subsystem, such as a combination of the UI and the vibration transaction. The purpose is to ensure that the subsystem's functions interact properly.

Both unit and integration testing will be performed using White Box testing Approach, which helps to detect bugs during implementation.

System Testing

In the final phase of implementation, we will integrate all subsystems into a comprehensive P2P vibration-based mobile payment system to test its functionality without faults in a controlled environment. It is a Black-Box Testing Approach to evaluate the system against defined use-case scenarios and to demonstrate usability results.

Performance Testing

We will test the accuracy of transaction transmission using the designed vibration-encode/decode algorithm under different scenarios. It helps validate and strengthen the design’s viability and quantify its reliability for further studies on the vibration channel in the mobile payment system.

Penetration Testing

Penetration testing (Pen test) is a simulated cyber attack designed to exploit vulnerabilities in a system before it is publicly released. It helps to validate if the theoretical cryptographic design is implemented securely in practice and achieves the security goals.

The testing results and discussion will be presented in Section 5.

3.5 Software Development Methodology and Stages

The project follows an Agile software development approach, with iterative development cycles. (see *figure 3.4*). Consider that the project has only a 6-month development period, from planning to application launch. The proposed SE approach is suitable for a rapid, incremental, and continuous delivery in a short-term period.

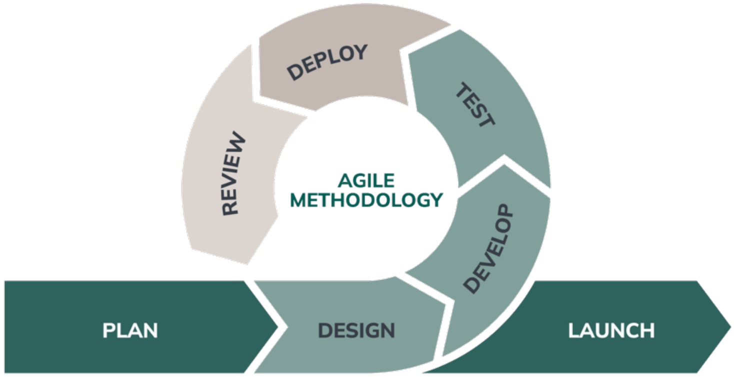

Figure 3.4: Diagram of the Agile software development cycle

The standard for this practice is that, after initial planning and goal, and use-case definition, we will start by designing a simple software application with limited functions. Next, we will begin developing another function and integrate it into the system by looping it several times, adding each function one at a time. Every time we add a new function, we will review and test its behaviour to ensure it is well-matched with the proposed system.

Therefore, the implementation has been distributed into five phases (1-5) to allow greater flexibility. The schedule is outlined below (More details about the high-level project schedule plan and Gantt Chart can be found in Appendix B):

a) Phase 1: P2P Payment System Interface (September 2025)
b) Phase 2: Basic Vibration channel development (October – November 2025)
c) Phase 3: Cryptographic framework development & Integration (December 2025 – January 2026)
d) Phase 4: System Integration & Testing (February 2026)
e) Phase 5: Final Preparation & Documentation (March 2026)

Figure 3.5 illustrates the agile development cycle for our project. We developed our system by focusing on small components in each iteration, integrating them one by one in each phase. Eventually, this approach will lead to the creation of our secure P2P payment system through vibration.

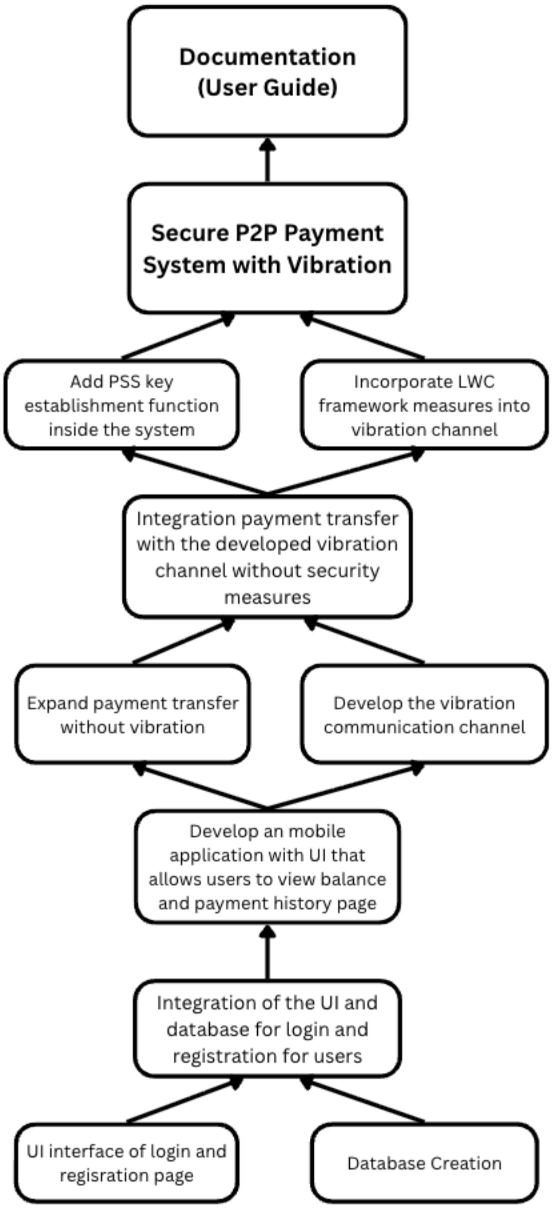

Figure 3.5: Agile software development cycle for the proposed system

4. Detailed Methodology and Implementation

This section provides a detailed implementation of the system's components, including its overall interfaces & database, the design of the vibration communication channel, and the Lightweight Cryptographic Protocol. Within each subsection, we will also discuss potential issues that may arise during development and provide guidance on how to address them.

4.1 User Interface Design

Based on the defined functionalities in Section 3.2, each function was implemented on different pages inside a mobile application. The interface design is intended to be simple and easy to navigate, with improved usability. As it is also a proof-of-concept application for the secure vibration channel, the UI design won't be the main focus of development. The P2P Vibration Mobile Payment System comprises five main pages: Login, Home, History, Transaction, and Management.

<u>Login Page</u>

This page allows users to log in to the payment system if they have an existing account in the database. They can create an account on the registration page if they are new users.

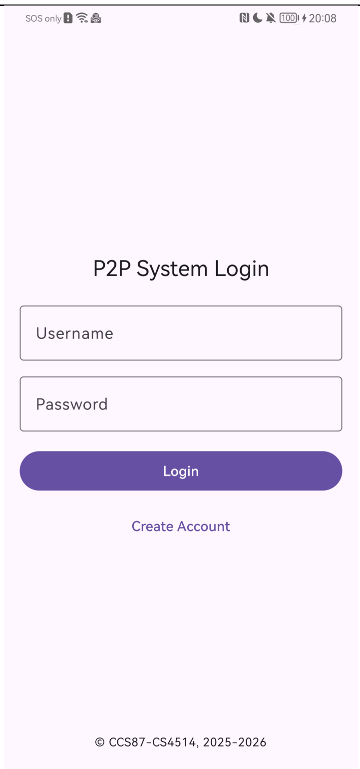

Figure 4.1: Layout of the login page

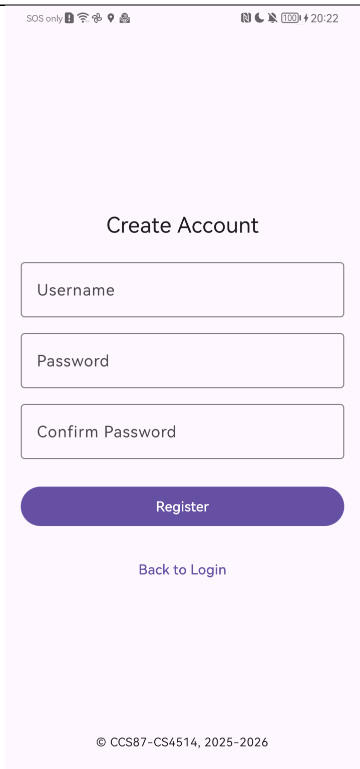

Figure 4.2: Layout of the Regisration page

Home Page

This is the main directory page, which contains all the functions that users can perform, allowing them to navigate to other pages, such as sending transactions or checking their payment history (See *Figure 4.3*).

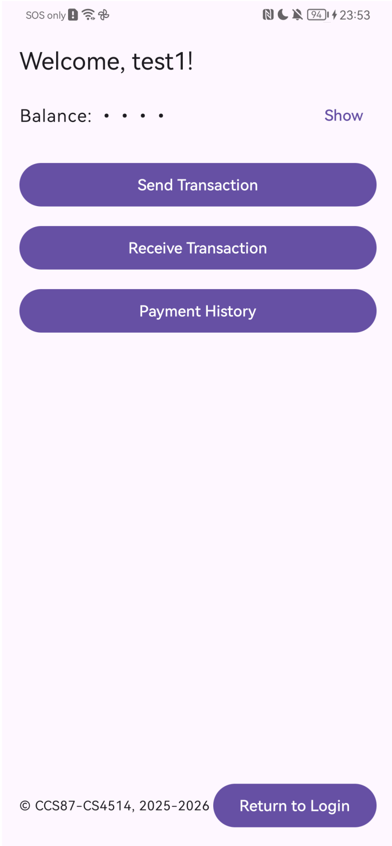

Figure 4.3: Layout of the User Home menu page

Payment History Page

On this page, users can view their transaction history, including payment and receipt records with other users. They can also search for the required transaction records by entering keywords in the search bar (See *Figure 4.4* below).

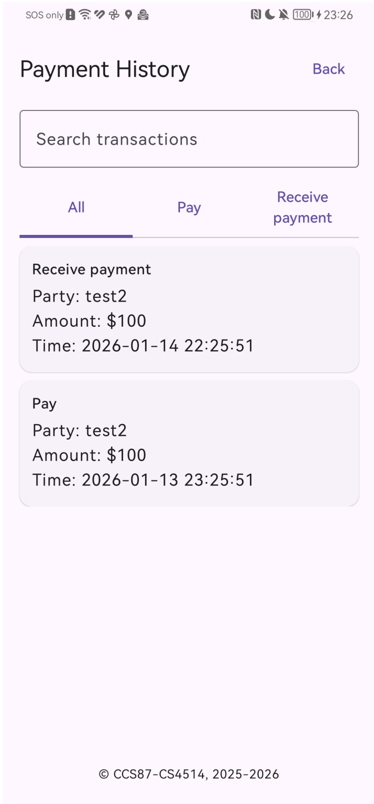

Figure 4.4: Layout of the payment history page

Transaction Page

The users can choose to transact $100 (“a”) or $200 (“b”), or receive payment from another user. Moreover, they must set up the PSS Key for encryption before the transaction. Detailed interaction will be shown in the figures below.

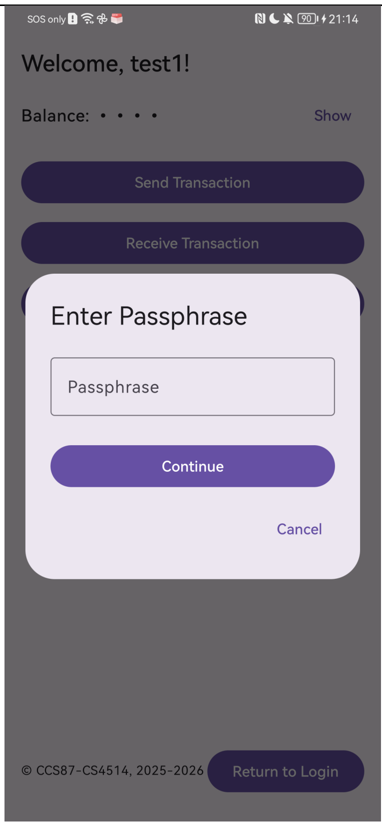

Figure 4.5: Passphrase input for PSS Key set up

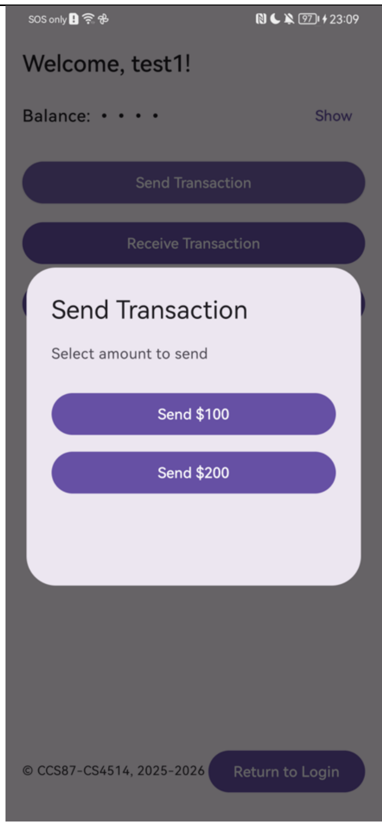

Figure 4.6: User can choose the payment amount

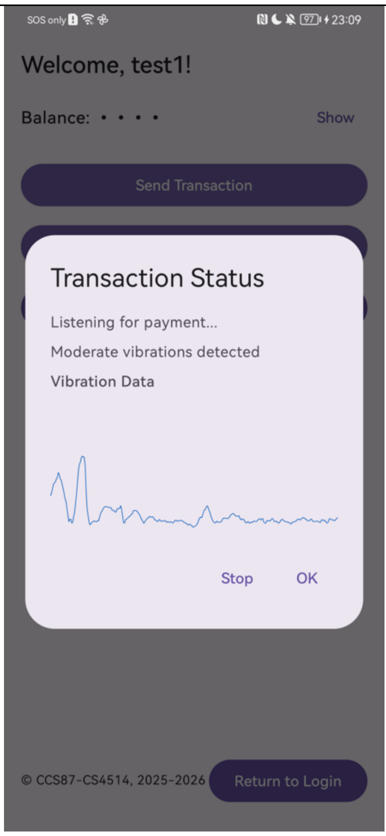

Figure 4.7: Transaction status – Vibration detection

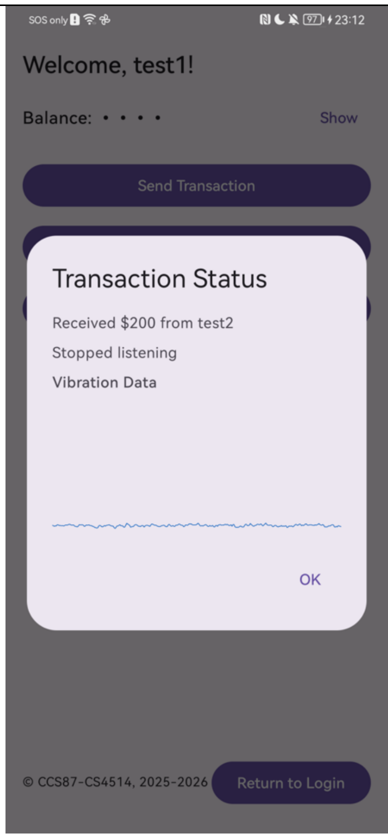

Figure 4.8: Completion transaction status

Admin Management Page

On this page, administrators can view the remaining balances of all users and manage their account statuses. Moreover, they can review all the transaction records users have previously performed.

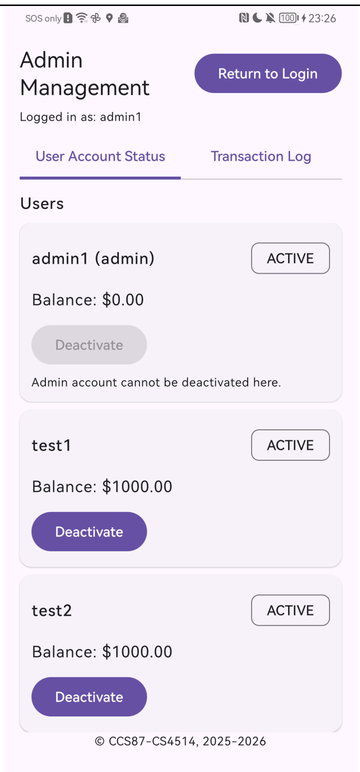

Figure 4.9: Layout for checking user status

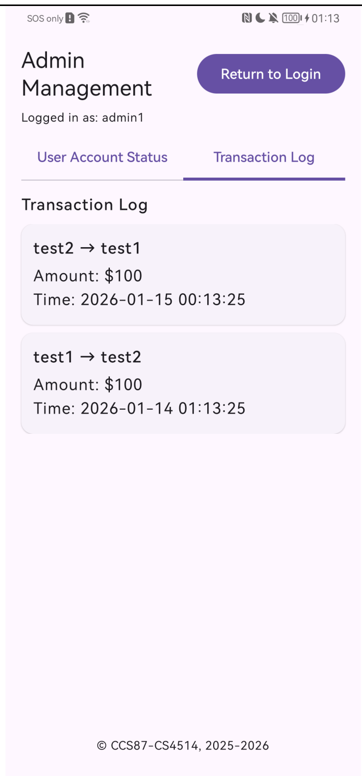

Figure 4.10: Layout for checking transaction log

4.2 Database Design

In our proposed P2P payment system, a simple database was implemented. As it is a proof-of-concept application for the secure vibration channel, the database also won't be the main focus of development. The ER diagram of the database is shown below.

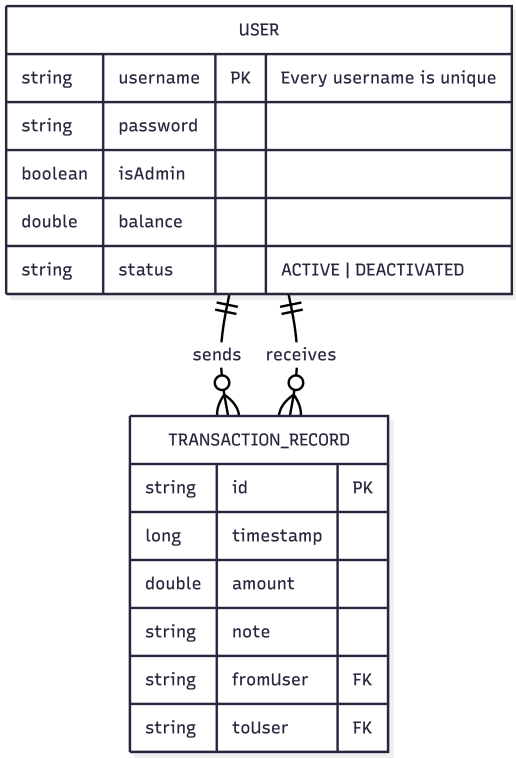

Figure 4.11: ER diagram of the database

4.3 Vibration Communication Channel Design

An overview of the design of vibration communication will be presented below. It is separated into two major components: 1) Vibration Encoder and 2) Vibration Decoder. The minor components that support the encoder and decoder: the Vibration Controller and Accelerometer Monitor, which use the built-in vibrator and accelerometer in the smartphones. *Figure 4.12* below shows the overall design of the vibration communication channel.

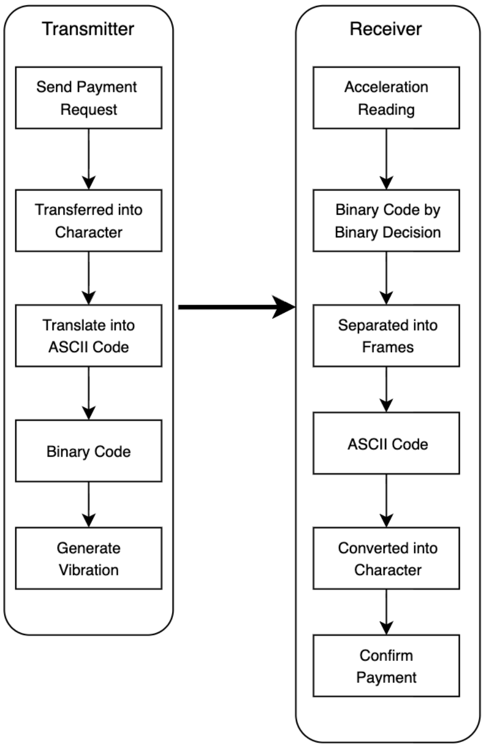

Figure 4.12: An overview of the vibration communication channel

The Vibration Controller is a simple and low-level hardware component that directly interacts with the mobile device's haptic motor. It is responsible for receiving a pre-defined vibration pattern from the encoder and executing it on the device vibrator. Moreover, it also ensures the vibration strength is set to maximum (255) only during vibration and "0" when it is paused.

The Accelerometer Monitor serves as the hardware interface for data collection and conversion. It detects the physical vibration by listening to the device's built-in accelerometer. It collects the raw sensor data and stores these readings in temporary buffers, then transfers the data to the decoder for processing. Below shows the details of the two key actions.

- Signal conversion: It collects the raw X, Y, Z-axis data from the accelerometer and converts it to a single-scale magnitude. The formula is defined as: $M = \sqrt{X^2 + Y^2 + Z^2}$.
- Data buffering: It helps to transfer all the object data to a list and pass it to the Vibration Decoder for analysis.

The Vibration Encoder is a primary component for the transmitter. The function of the encoder is to transfer a payment request into the vibration pattern. Below shows the implementation process:

- Data Framing: The encoder first converts the payment request (Data) into characters, such as "a" corresponding to \$100 and "b" corresponding to \$200, and then converts it to a 7-bit ASCII code. Then, the encoder will construct the vibration pattern into a frame format of "Start (00010) + Data (1100001) + End (00011)", resulting in a total of 17-bit data frame.

- Modulation: The data frame will then be modulated using the OOK scheme. Each bit is allocated into a 1-second time slot, and the data rate is fixed at 1 bit per second (bps). Where bit '0' is a 1000ms pause (No vibration), and bit '1' is a 200ms vibration, and padded with 400ms on either side (400ms pause + 200ms vibrate + 400ms pause) to fit the 1000ms window.

Table 4.1 shows a simple illustration of the encoding algorithm:

Algorithm 1 Vibration encoding algorithm

Require: Command character C
Ensure: Data Frame D

```
asciiCode ← ASCII(C)
binaryData ← ToBinary(asciiCode, length = 7)
startMarker ← "00010"
endMarker ← "00011"
fullBinary ← startMarker || binaryString || endMarker
D ← ∅
for all bit in fullBinary do
    if bit == '1' then
        Append 400ms to D
        Append 200ms to D
        Append 400ms to D
    else
        Append 1000ms to D
    end if
end for
return D
```

Table 4.1: Pseudocode of the encoding algorithm

The results are a 17-second vibration sequence, such as [1000, 1000, 1000, 400, 200, 400, ...], which is then passed to the Vibration Controller for execution. *Figure 4.13* illustrates the process that occurs after users select the payment amount.

```
Encoding command 'a' (ASCII: 97)
Full binary: 00010110000100011 (17 bits)
Generated pattern: 1000, 1000, 1000, 400, 200, 400, 1000, 400, 200, 400, 400, 200, 400, 1000, 1000, 1000, 1000, 400, 200, 400, 1000, 1000, 1000, 400, 200, 400, 40
Pattern entries: 29
Total duration: 17000ms (17.0s)
Pattern duration: 17000ms
Pattern: 1000, 1000, 1000, 400, 200, 400, 1000, 400, 200, 400, 400, 200, 400, 1000, 1000, 1000, 1000, 400, 200, 400, 1000, 1000, 1000, 400, 200, 400, 400, 200, 40
Amplitudes: 0, 0, 0, 0, 255, 0, 0, 0, 255, 0, 0, 255, 0, 0, 0, 0, 0, 0, 255, 0, 0, 0, 0, 0, 255, 0, 0, 255, 0
Vibration started successfully
```

Figure 4.13: Logs when encoding the payment command

The Vibration Decoder is a primary component for the Receiver. It analyzes the raw sensor data collected by the Accelerometer Monitor to reconstruct the original data. This process is triggered automatically after a fixed buffering period and involves two main stages.

- Bit Reconstruction: A function is available for converting analog accelerometer data back into a digital bitstream. It iterates through the buffered list (Depending on the size of the transaction) in a discrete 1-second window corresponding to the synchronous bit rate. For each window, it checks whether the calculated magnitude exceeds a set threshold (required magnitude) a sufficient number of times. If vibration is detected, a "1" is appended to the binary string and vice versa. For the threshold and required count, these values can be adjusted to any desired level based on specific needs.

- Pattern Matching: The constructed binary string will be passed to another function that searches for the 17-bit data frame to verify its validity. This is achieved by finding the start marker [i, …, i+5] and subsequently checking the 12 bits for the end marker [i+12, … i+17]. The data frame in the middle is then extracted for processing.

Table 4.2 shows a simple illustration of the decoding algorithm:

Algorithm 2 Vibration Decoding Algorithm

```
Require: List of Accelerometer datas A, Threshold τ = 9.65
Ensure: Decoded Commands C
  Stage 1: Bit Reconstruction
  Bits ← ∅
  T_bit ← 1000ms
  for each time window W of duration T_bit in A do
    count ← Number of samples in W where value > τ
    maxVal ← Maximum sample value in W
    if count > 5 AND maxVal > τ then
      Append '1' to Bits
    else
      Append '0' to Bits
    end if
    if Length of Bits ≥ 17 then
      break
    end if
  end for
  Stage 2: Frame Matching
  StartMarker ← "00010"
  EndMarker ← "00011"
  C ← ∅
  for i ← 0 to Length(Bits) do
    window ← Bits[i ... i + 17]
    if window starts with StartMarker AND ends with EndMarker then
      DataFrame ← window[5 ... 11]
      AsciiVal ← BinaryToDecimal(DataFrame)
      Command ← Char(AsciiVal)
      if Command ∈ {'a','b'} then
        Add Command to C
      end if
    end if
  end for
  return C
```

Table 4.2: Pseudocode of the decoding algorithm

Finally, if only a single valid pattern is found, the decoder converts the ASCII value back and identifies it as the payment command to process the required transaction. If multiple valid patterns are detected due to signal noise, the user is presented with a list of possibilities to help them make the final selection. The figures below illustrate the process that occurs behind the decode algorithm during receiver listening to the vibration patterns.

```
Data: 8400 points, Max: 9.73, Vibrations: 0
*** HIGH VIBRATION: 9.70 at 16812ms ***
*** HIGH VIBRATION: 9.68 at 16822ms ***
*** HIGH VIBRATION: 9.67 at 16836ms ***
*** HIGH VIBRATION: 9.69 at 16845ms ***
*** HIGH VIBRATION: 9.68 at 16856ms ***
Data: 8500 points, Max: 9.70, Vibrations: 0
Data: 8600 points, Max: 9.61, Vibrations: 0
Data: 8700 points, Max: 9.61, Vibrations: 0
Data: 8800 points, Max: 9.61, Vibrations: 0
Data: 8900 points, Max: 9.61, Vibrations: 0
Data: 9000 points, Max: 9.61, Vibrations: 0
```

Figure 4.14: Detection of Vibration during Listening process

```
BIT DETECTION: Duration=20113ms, Expected bits=20
Bit 0: max=9.63, avg=9.60, vibrations=0 -> 0
Bit 1: max=9.62, avg=9.60, vibrations=0 -> 0
Bit 2: max=9.63, avg=9.60, vibrations=0 -> 0
Bit 3: max=9.74, avg=9.60, vibrations=14 -> 1
Bit 4: max=9.62, avg=9.59, vibrations=0 -> 0
Bit 5: max=9.71, avg=9.59, vibrations=22 -> 1
Bit 6: max=9.71, avg=9.59, vibrations=21 -> 1
Bit 7: max=9.61, avg=9.59, vibrations=0 -> 0
Bit 8: max=9.61, avg=9.59, vibrations=0 -> 0
Bit 9: max=9.61, avg=9.59, vibrations=0 -> 0
Bit 10: max=9.61, avg=9.59, vibrations=0 -> 0
Bit 11: max=9.75, avg=9.59, vibrations=25 -> 1
Bit 12: max=9.60, avg=9.59, vibrations=0 -> 0
Bit 13: max=9.61, avg=9.59, vibrations=0 -> 0
Bit 14: max=9.61, avg=9.59, vibrations=0 -> 0
Bit 15: max=9.74, avg=9.59, vibrations=26 -> 1
Bit 16: max=9.73, avg=9.59, vibrations=20 -> 1
COMPLETED BIT DETECTION: 17 bits
RAW BINARY DETECTED: 00010110000100011 (17 bits)
FINAL DECODE: Searching for pattern in 17 bits
PATTERN FOUND at position 0: start=00010, data=1100001, end=00011
VALID COMMAND: 'a' (binary: 1100001, ASCII: 97)
SUCCESS: Single command 'a' decoded
```

Figure 4.15: Logs when decoding the payment command

During implementation, due to time delays of starting the transaction, the decoding algorithm may sometimes incorrectly decode the command, for instance, Sometimes when the decoder tries to decode the character "b," it may confuse it with the character "c," since the 7-bits binary representations of both characters are similar: 110001**0** ("b") and 110001**1** ("c"). The algorithm may incorrectly recognize the last bit as "1" but not "0" and make a wrong decision. The reason for this fault is probably that the algorithm recognizes the high vibration in the previous window in the current window, which misleads the last bit. To solve the problem, we have to repeatedly test the algorithm and carefully adjust the threshold and sample count, so that even when some samples are included in the wrong windows, they won't meet the algorithm's condition and won't be recognized as "1".

4.4 Lightweight Cryptographic Protocol Design

The vibration communication channel, as described in Section 4.2, provides a novel method for data transmission. However, the pure data frames are transmitted without any protection, making them may vulnerable to several attacks and failing to meet the security goals of the mobile payment system. This section provides the details of the LWC design to secure the channel. To prevent potential attacks and fulfill the security goals outlined in the literature review (Section 2.3.1), we should implement a simple cryptographic algorithm to secure the communication channel.

As described in the literature review section (Section 2.3.2). Since the vibration channel is a low-bandwidth (1 bps) channel, we should consider using the lightweight cryptographic frameworks stated in section 2.3.3. Symmetric key cryptography should be used. We should also consider how and which protocol should be implemented in a lightweight method.

Based on the findings of the literature review (Section 2.3.3), the framework is constructed based on three standard lightweight symmetric protocols listed below:

- Key Derivation: To solve the key exchange problem of the symmetric encryption of AES, both the transmitter and receiver will first verbally agree on a shared passphrase. After exchanging the passphrase, the system will process it using PBKDF2 with a static salt and a fixed iteration count. The hard-coded method ensures that both devices generate the same 128-bit PSS key. The PSS key will be stored only temporarily in the storage and will be removed after the transaction is complete, and it is not expected to be reused.
- Encryption Scheme: Considering the transmission time of the vibration channel, we will consider using AES-CTR. This converts the AES block cipher into a stream cipher, enabling it to generate a keystream of the exact length required to encrypt the data, thereby avoiding padding that could result in increased transaction time.
- Integrity and authentication: To prevent tampering, a HMAC-SHA256 is used. The 256-bit output is truncated to 16-bit tags and concatenated into the data.

Below, this project proposes a new lightweight cryptographic framework with the protocols reviewed. The notations used to describe the protocol are defined in *Table 4.3* below.

Notation	Description
Passphrase	An out-of-band, secret shared passphrase between the two users
PSS_Key	Pre-Shared Secret key
PBKDF2	Password-Based Key Derivation Function version 2
M	Original data frame of the payment message
P	Plaintext combines the message and the tag
C	Ciphertext that encrypted the plaintext
Payload	The final data stream transmitted via vibration
N	Nonce
T	The authentication Tag

K	The pseudorandom Keystream generated by AES-CTR
HMAC	Hash-based Message Authentication Code
truncate	A standard function that shortens a binary bitstring
AES_CTR	AES in Counter mode
$\oplus$	The bitwise XOR (Exclusive OR) operation
$\|\|$	A concatenation operation

Table 4.3: Notations

The original 17-bit data will not be transmitted at first. It is encrypted and wrapped into a new secure data. The new total data transmitted over the vibration channel is 41 bits, including the security measures. *Figure 4.16* below shows the new data structure after encryption, which will be sent to the receiver and the details of the modified plaintext.

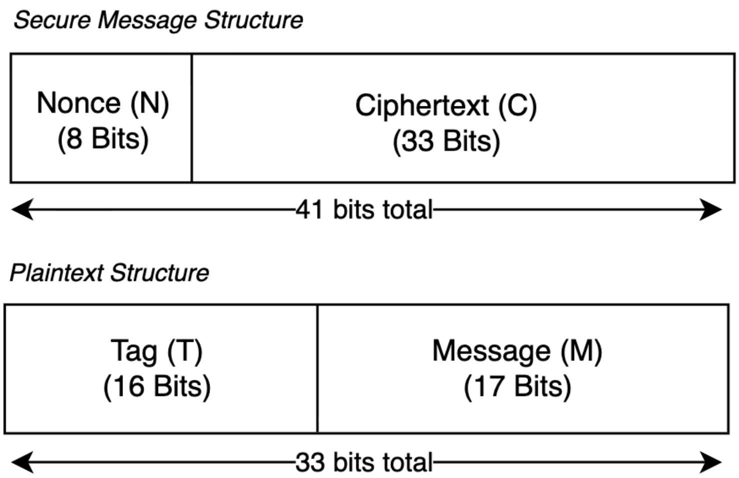

Figure 4.16: Final Data Stream Structure

An 8-bit nonce (N) is a random initialization vector included in the message to ensure that every transaction is unique and distinct. The system will generate a random number from 0 to 255 and get the 8-bit binary string as the nonce to fit the design of the OOK scheme for the vibration channel. Inside the 33-bit plaintext, it includes a 16-bit HMAC tag (T), which is truncated from HMAC-SHA256, and the original 17-bit vibration message (M).

A general workflow of the proposed schemes is detailed as follows:

First step: Both the sender and Receiver verbally agree on a passphrase, and the system will derive an identical 128-bit PSS Key for both users.

- $PSS\_Key = PBKDF2(passphrase)$

Second step: The sender will first construct a 17-bit message for the payment request and generate a unique 8-bit nonce. Then, the system will generate a 16-bit HMAC integrity tag using the message, nonce, and the PSS key, and also generate a 33-bit keystream with AES-CTR using the PSS key and Nonce. Next, the system combines the message and the tag into plaintext and encrypts it using the keystream to produce the ciphertext by the XOR operation. Finally, the system assembles the nonce and ciphertext into the 41-bit payload and passes it to the vibration controller to send out.

- $T = truncate(HMAC(PSS\_Key,\ M \,||\, N), 16\ bits)$
- $K = AES\_CTR(PSS\_Key, N)$
- $P = M \,||\, T$
- $C = P \oplus K$
- $Payload = N \,||\, C$

Third step: It is just a reverse step of the second step, where the VibrationDecoder reconstructs the 41-bit stream from the vibrations and splits the received payload into an 8-bit nonce and a 33-bit ciphertext. Then the system uses the PSS key and the received nonce to generate the same keystream and decrypt the ciphertext back to the plaintext. Next, the system will split the 33-bit plaintext into the received 16-bit tag and the 17-bit messages, and calculate its own expected tag for verification. If the expected tag matches the received tag, then it indicates the authentication and integrity of the data, and the transaction will be processed. Otherwise, the transaction will be terminated by the protocol.

- $K = AES\_CTR(PSS\_Key, N')$
- $P' = C' \oplus K$
- $T' = truncate(HMAC(PSS\_Key,\ M' \,||\, N'), 16\ bits)$

Figure 4.17 below shows the description of the proposed protocol.

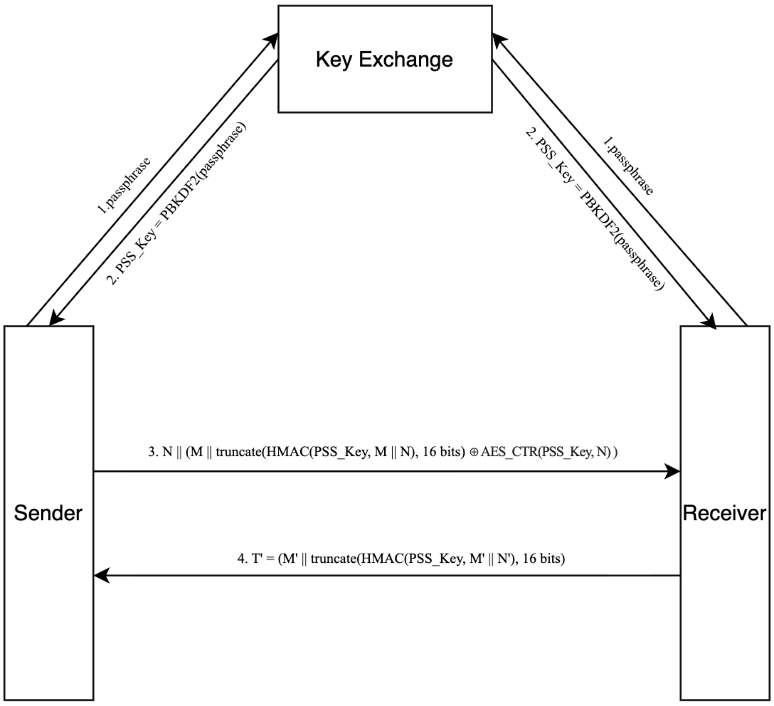

Figure 4.17: Description of the proposed protocol

The figures below show one typical operation performed behind LWC during sending and receiving vibration transactions. *Figure 4.18* shows how the PSS key and encrypted message are established from the sender, and *Figure 4.19* demonstrates how the receiver decrypts the payload and decodes the vibration patterns into the payment command.

```
Key derivation requested
Passphrase length=4
Salt (static) length=16 bytes
Derived PSS key length=16 bytes
Derived PSS key (hex)=135beb7a8637e996c398e752200be795
=== Secure Sender Build Start ===
M (17-bit)=00010110001000011
Nonce N (0..255)=111
N (8-bit)=01101111
T (16-bit, truncated HMAC-SHA256)=1100110111100100
P = M || T (33-bit)=000101100010000111100110111100100
K (33-bit keystream via AES-CTR)=100101001000101111111001010110000
C = P XOR K (33-bit)=100000101010101000011111101010100
Payload = N || C (41-bit)=01101111100000101010101000011111101010100
=== Secure Sender Build End ===
```

Figure 4.18: Establishment of the PSS key and secure payload

```
[38887] COMPLETED BIT DETECTION: 41 bits
[38887] RAW BINARY DETECTED: 01101111100000101010101000011111101010100 (41 bits)
rawBits(len=41)=01101111100000101010101000011111101010100
[38887] SECURE RX: payloadBits41=01101111100000101010101000011111101010100
[38887] SECURE RX: N(bits)=01101111 N(int)=111
[38887] SECURE RX: C(33)=100000101010101000011111101010100
secureRx payload41=01101111100000101010101000011111101010100
secureRx nonceBits8=01101111 nonce=111
secureRx ciphertext33=100000101010101000011111101010100
[38888] SECURE RX: K(33)=100101001000101111111001010110000
[38889] SECURE RX: P'(33)=C XOR K =000101100010000111100110111100100
secureRx keystream33=100101001000101111111001010110000
secureRx plaintext33=000101100010000111100110111100100
[38889] SECURE RX: M'(17)=00010110001000011
[38889] SECURE RX: T_recv(16)=1100110111100100
[38889] SECURE RX: T_exp(16)=1100110111100100
secureRx tagRecv=1100110111100100 tagExp=1100110111100100
[38889] SECURE RX: TAG OK → decode legacy 17-bit message
[38889] FINAL DECODE: Searching for pattern in 17 bits
[38889] FOUND COMMAND: 'b' at index=0 (ASCII=98)
[38889] SUCCESS: Single command 'b' decoded
disableSensor handle:0, packageName:com.example.p2p_system.VibrationDecoder
```

Figure 4.19: Procedures of the decryption scheme

5. Preliminary Results and Future Improvement

As mentioned before, this project addresses the limitations of using the vibration channel in the offline P2P payment system and designs a lightweight cryptographic protocol for this channel to meet security goals. In this section, we will present the preliminary findings from the system's functional, performance and penetration testing. Moreover, we will discuss the possible reasons for each outcome and whether each outcome fulfilled the project goals or identified parts of the system for further improvement.

5.1 System Functional Testing Results

As mentioned in section 3.4, the results will be illustrated by presenting functional test cases for different scenarios that may be encountered when using the application. The primary objective of this phase was to validate the functional logic of the proposed secure P2P vibration payment system. *Figure 5.1* shows the controlled test environment setup, where the sender and receiver are closely attached to each other and are placed on a stable wooden table. Appendix C presents the use cases performed during system functional testing and the final results of each test.

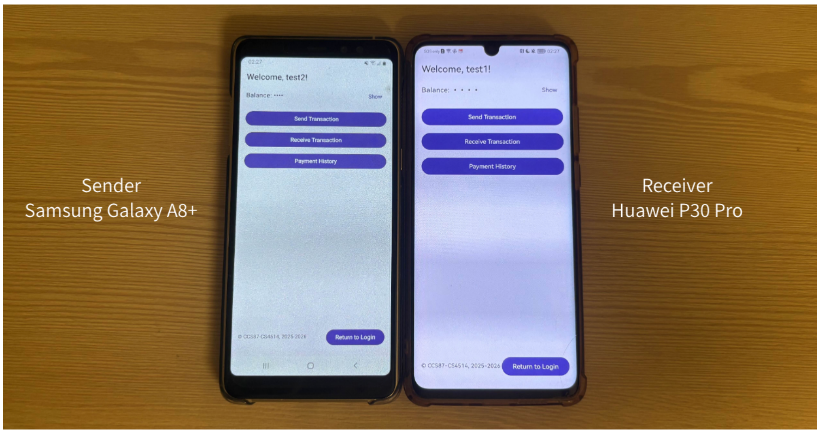

Figure 5.1: Users' test environment setup

Overall, the successful results of all use cases demonstrated that the P2P payment system of the software-to-hardware pipeline is correctly implemented. From the usability perspective, it shows that a user can perform the application as a financial application. The sender can encode and encrypt the transaction request into a vibration pattern, and the receiver recognizes and decrypts it to the payment command. This proves that the proposed vibration communication channel and lightweight cryptography protocol are logically functional and usable in a secure P2P payment system background.

However, we should note that these tests were conducted at 0 cm, which represents the "ideal environment" where vibration energy loss is minimal. They have not yet proven the system's resilience to external variables such as communication distance or surface materials. Moreover, we did not test the performance of each magnitude threshold of our decode algorithm. It indicated that there are areas for further investigation to prove that it is a comprehensive secure P2P payment system.

5.2 Performance Testing Results

To address the issues stated in the previous subsection, we also test the accuracy of the designed vibration-encode and decode algorithm at different adaptive magnitude thresholds, communication distances, and surface materials, both with and without the cryptographic framework. The sender and receiver are still the same mobile phones presented in section 5.1.

5.2.1 Performance based on Magnitude-Based Threshold

Vibration detection is tested with different thresholds by varying the magnitude required to be recognised as vibration, since the idle-tested phone magnitude ranges from 9.59 to 9.61. Therefore, the tested magnitude thresholds should be higher than the range value, such as 9.65, 9.75, 9.8, 9.9, and 10. The accuracy of the received data reflects its correctness.

Threshold	Without LWC Framework	With LWC Framework
9.65	>95%	>95%
9.75	>95%	>95%
9.8	~77%	~54%
9.9	~65%	~49%
10	~65%	~36%

Table 5.1: Performance based on magnitude threshold

Table 5.1 shows the results. It can be observed that accuracy decreases significantly after the threshold exceeds 9.75; it continues to drop thereafter, and the transaction will fail. Furthermore, we spotted that the accuracy without the LWC framework was fixed around 65%, whereas with the LWC framework it performed poorly and unstably. The possible reason for these interesting results is that the transaction without the LWC framework generated the same vibration pattern according to the defined algorithm; the results are always the same, and the receiver may not be able to map the same bit in every attempt. But if the algorithm uses the LWC framework, since the nonce introduces randomness and generates different vibration patterns in every transaction after encryption, the structure is more complex, leading to vibration data that may not map to the receiver's arbitrary state. However, if the threshold is within 9.65-9.75, the accuracy remains high and stable, and the transaction will succeed regardless of whether the LWC framework is used. It shows that the ideal threshold for the transaction lies between 9.65 and 9.75.

5.2.2 Performance based on Medium Objects

Different types of media used for transactions may also affect accuracy. We tested our vibration channel on four types of common medium objects. The experimental setup is also similar to section 5.1, but the placement locations are different, and the same ideal threshold is used.

Materials	Without LWC Framework	With LWC Framework
Wood Table	>95%	>95%
Metal Shelf	>95%	>95%
Cushioned chair	~85%	~68%
Cotton bed	~65%	~51%

Table 5.2: Performance based on surface materials

From the results shown in Table 5.2, we can see that the transaction on the wood table and the metal shelf have higher accuracy, while the cushioned chair and cotton bed have lower accuracy and result in failed transactions. The possible reason for this result is the material density. Wood and table are more solid and high-density, with low acoustic damping, resulting in better performance. However, the cushioned chair and cotton are more porous; the soft materials may absorb and weaken high-frequency vibration, making it difficult for the receiver to decode the data.

5.2.3 Performance based on Communication Distance

The communication distance may also affect the accuracy, as it is a close-proximity channel; the vibration strength may get weaker for the receiver to detect. We should also test the data reception at distances of 10cm, 20cm, 30cm, and 40cm.

Distance	Without LWC Framework	With LWC Framework
10cm	>95%	>95%
20cm	>95%	>95%
30cm	~65%	~41%

40cm	~65%	~32%

Table 5.3: Performance based on transaction distance

Table 5.3 presents the testing results. We saw that accuracy becomes low and fixed after exceeding 30cm, indicating that the receiver could not receive all the messages correctly beyond that distance, and all transactions will fail thereafter. It has also been shown that the effective range of the transaction may be within 20cm, and our proposed secure vibration channel is robust for transferring vibration patterns within this range.

5.2.4 Discussion Based on Performance Results

To conclude, results from section 5.2 have demonstrated that our proposed secure P2P payment system using the vibration channel may be viable and reliable under certain circumstances, such as when the vibration algorithm is under an ideal attitude threshold, it is placed on a stable medium object for transaction, and in a short communication distance.

Nevertheless, the LWC framework performs worse and is less stable than the one without it across all results. This is possibly because the structure of the payload required to transfer increased; it is suggested that we can improve the current decoding algorithm by introducing more techniques, such as adaptive amplification and frequency to improve the proposed LWC framework accuracy.

Furthermore, although we have tested our algorithm based on the communication distance, each test was performed only in the same horizontal direction, and we haven't evaluated it from other communication directions. It can be further tested, as different angles of view may also affect bitwise accuracy.

5.3 Penetration Testing Results

The pen test will focus on the security of the vibration communication channel and the key establishment. Since the key is used to encrypt the vibration data in each transaction, it is expected that the key will not be reused. Hence, the ethical hacking attack is expected to mainly target the proposed security design to test whether it is vulnerable to attacks such as replay attacks and eavesdropping.

First, to address key reuse, our implementation of the LWC protocol uses the clearkey function after the sender and receiver have completed the transaction, the transaction timed out, or the transaction was cancelled (see *Figure 5.2* below). It ensures the key won't be reused and remains unique for encryption.

```
11 Usages
fun clearKey(key: ByteArray?): ByteArray? {
    if (key == null) return null
    for (i in key.indices) key[i] = 0
    return null
}
```

Figure 5.2: ClearKey function

Next, it is essential to examine the security challenges and highlight the necessity of a cryptographic design. First, it may be vulnerable to eavesdropping if the attacker is also in physical proximity and could intercept the vibrations, decode the messages, and discover the payment request from the users. Next, it is possible to replay attacks, as an attacker can record a valid vibration pattern and replay it later to authorize a duplicate payment.

In our design, the sender and receiver share a passphrase to drive a unique 16-byte PSS key, then the system generates a 33-bit keystream using AES in Counter (CTR) mode with the PSS key and an 8-bit unique nonce. Lastly, the 17-bit payment message is combined with a 16-bit HMAC tag and then XORed with the keystream; a 44-bit payload is sent and received. Consequently, the security results show that even if the attacker is in physical proximity and intercepts the vibration pattern, they cannot decode our payload without the secret PSS key, which only the original sender and receiver know, keeping the payment details confidential. Moreover, if the attacker records and replays the old vibration message, the receiver will detect that the nonce or the resulting HMAC tag is no longer valid and terminate the transaction. The double protocol protects the transaction integrity. Therefore, from a theoretical perspective, our protocols may help protect the transaction and fulfill the security goals for a P2P payment system.

Moreover, the results of the performance-based test on communication distance indicate that the scammer needs to be at least 20cm away to eavesdrop. The proposed secure transaction channel ensures that two users are in close proximity and that attacks are highly impossible, as users can easily discover third parties and identify the sender and receiver in a real-life setting, thus guaranteeing the security of the payment channel.

Lastly, we identify an area that may require improvement; the current protocol may not be secure from an availability perspective, as the attack could play a random vibration pattern and disrupt the transaction process. The project may be able to carry out a collision-detection mechanism in the decode algorithm to bypass interference from unknown sources.

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Appendices

Appendix A - Monthly Log

October 2025

I will provide my progress report in three sections each month.

It includes: 1) Documentation 2) Implementation 3) Expectations/ goals for the next month.

For Documentation, I have reviewed the comments on the project plan from my supervisor and have completed the draft of the introduction and literature review part for the interim report I this month. Moreover, I have brainstormed the design and testing methods for the overall system.

For implementation, I have completed the UI coding part of the P2P payment system. However, I'm kind of behind schedule due to the busy midterm weeks and assignments. For the vibration channel part, I have completed the encoding scheme to generate vibration. However, I am still stuck on a minor issue related to detecting vibration by calculating magnitude data for the decoding scheme.

Expectations to be complete next month:

- Complete the draft of the proposed design and implementation/testing schedule of the interim report 1 and submit it to my supervisor for suggestions and reviews
- Complete basic vibration communication channel code development (Without encryption)
- Study on proposed lightweight cryptographic protocols and consider how to integrate them into the communication channel

November 2025

For Documentation, I have completed the remain part of the interim report I and submitted to my supervisor. We have also scheduled a meeting to talk about the project. He have provided some suggestion on the implementation method of the vibration design and solved my questions.

For implementation, I have completed to remain decoding part and start considering how to integrate the cryptographic protocols into the communication channel.

Expectations to be complete next month:

- Study on the proposed lightweight cryptographic protocols and start implementing it into the vibration communication channel
- Check the comment from my supervisor on interim report I and start consider how to write interm report II

December 2025

For Documentation, I reviewed the comments from my supervisor and modified the requirements that he required me to change and started writing my implementation details on the interim report II
For implementation, I have completed the decoding part; however, some issues have been identified. I have found some time to address the issue, and now consider the design of incorporating security protocols into it.

Expectations to be complete next month:

- Finish the proposed lightweight cryptographic protocols in the vibration communication channel
- Extract the initial testing results on the design and complete the interim report II

January 2026

For implementation, after some minor fix of UI and vibration communication channel, I have successfully integrated the purposed cryptography inside the P2P payment system, I'm currently evaluating the developed system from test cases and performances and extract the initial testing results.

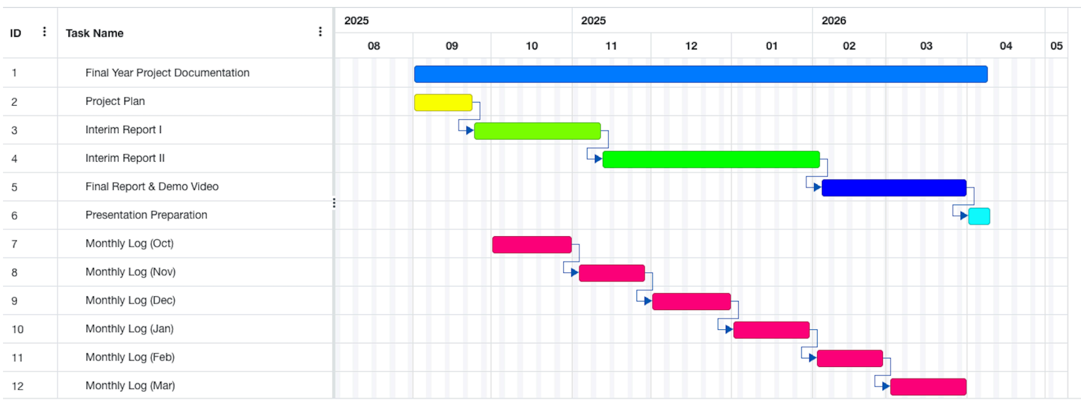
For Documentation, I have completed the interim report 2, and send to my supervisor for reviewing. I'm waiting for his comments and prepare for the final submissions.

Expectations to be complete next month:
- Improve the quality of the sources code and identify any hidden bugs
- Continue testing the developed system
- Preparation for the final report and submission required

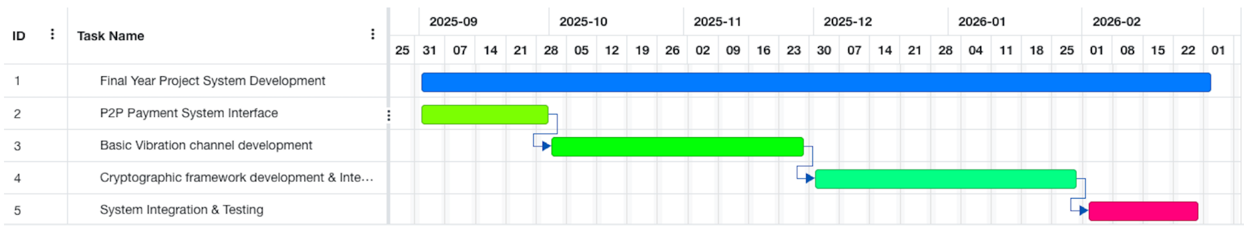
Appendix B - High-level project schedule plan & Gantt Chart

Goals	Key Deliverables	Improvement on the previous phase	Testing	Documentation
Phase 1: P2P Payment System Interface (September 2025)				
Develop a P2P payment system interface for two Android phones	A functional payment interface with basic transaction operations	N/A	Unit testing on UI interfaces	Project Plan
Phase 2: Basic Vibration channel development (October – November 2025)				
Set up the development environment and implement core vibration transmission and reception between two phones	A basic Android app that can encode a payment command into a vibration frame and decode it back on the receiver device	Implementing vibration and payment request into the UI interface	Unit and integration testing on the UI interfaces and the vibration communication channel	Interm Report I
Phase 3: Cryptographic framework development & Integration (December 2025 – January 2026)				
Design and develop the passphrase-based key exchange and AES	An Android app prototype that can perform a secure key exchange and send	Integrating the proposed secure protocols into the vibration	Unit and integration testing on the UI interfaces and the vibration	Interm Report II preparation

encryption with the HMAC scheme	an encrypted message via vibration	communication channel and UI interfaces	communication with secure cryptographic protocols	
Phase 4: System Integration & Testing (February 2026)				
Integrate all components into a single, comprehensive system and conduct testing and evaluation of the system	A fully functional end-to-end secure P2P vibration mobile payment application	Improve the readability and quality of the source code	System testing on the whole system, and perform preliminary performance and Penetration testing	Interm Report II & Final Report preparation
Phase 5: Final Preparation & Documentation (March 2026)				
Prepare the final report, create a demo video, and fine-tune the source code for final submission. Moreover, preparation for the FYP presentation.	Final project report, complete source code, demonstration video, and a presentation slide	Improve the readability and quality of the source code	Final performance and penetration testing	Same as Key Deliverables



A Gantt Chart for the Documentation Part of the Project



A Gantt Chart for the System Development part of the Project

Appendix C - Use cases and the final functional testing results

User Create Account	
Requirement:User type the username and a password longer than 6 character for register account	
Test Case:	Expected Result:
User input a valid username and valid password Username is not being registered	User created an account in local database and forward back to the login page for login
User input a valid username and valid password But the username is being registered	Alert ("Username is being registered, please use another name!") Stay in register page
User input a invalid username (Exceed String Limit or too short) or invalid password that break the security rule	Alert ("Invalid username, Please try again!") or Alert ("Invalid password, Please try again!") Stay in register page
Final Results: Success	

User login in account	
Requirement:User type the valid username and password to login in	
Test Case:	Expected Result:
User input a valid username and valid password (Account is existed)	User Logged in, forward user to Home menu display welcome message User data retrieved from local database
User input a invalid username and valid password	Alert ("Please input an valid username!") Stay in login page
User input a valid username and invalid password	Alert ("Invalid password, Please try again!") Stay in login page
User input invalid username or password multiple time	Alert ("Login in function locked, please try again later!") Back to mobile home page
Final Results: Success	

Checking payment history	
Requirement:User want to check transaction record and enter payment history page	
Test Case:	Expected Result:
User has payment record	Automatically display the transaction record in the page
User has no payment record	No payment record in the page
Final Results: Success	

Admin Management page	
Requirement: Administrator want to check the transaction records and users account status	
Test Case:	Expected Result:
Admin click “User Account Status”	Display all the user (Including admin) in the list
Admin click “Transaction Log”	Display all user transaction record
Final Results: Success	

PSS Key generation	
Requirement: User’s establish the share key using passphrase before performing transaction (Sender and Receiver)	
Test Case:	Expected Result:
Both user’s enter the exact same secret passphrase	Both devices process the passphrase and generate the identical PSS key for processing transaction later
One of the user enter the different secret passphrase	The generated PSS key will be different, transaction will fail alert ("Authentication Failed!" or "Corrupt Data!") during transaction process
Final Results: Sucess	

Send Payment Transaction	
Requirement: Sender send payment transaction to receiver (Assume PSS key is successfully establish for both users and place phones on the same surface)	
Test Case:	Expected Result:
Tap “Send Payment”	The payment request will be encrypted and the vibration encoder will encode the data based on the OOK algorithm to vibration pattern. The phone will physically vibrates
User tap “Cancel payment request” after requesting payment	The transaction request will be terminated, no money will be paid
Final Results: Success	

Receive Payment Transaction	
Requirement: Receiver receive payment transaction to sender (Assume PSS key is successfully establish for both users and place phones on the same surface)	
Test Case:	Expected Result:
Tap “Receive Payment” (HMAC Tag same)	The vibration decoder will listen and capture the vibration pattern and and decodes it using the adaptive threshold algorithm, decrypt the data, calculate the expected tag and confirms the payment
Tap “Receive Payment” (HMAC Tag not the same)	After calculatd the expected tag but it is not same with the received tag, transaction will fail and terminated, alert ("Authentication Failed!")

User tap "Cancel payment receive" after receive payment	The listen of vibration will be terminated, no money will be receive
Final Results: Success	